

Confidence Interval Construction: Examples

Peter E. Freeman - Statistics & Data Science - Carnegie Mellon University

June 2024

The Setup

Let's assume that we sample n independent and identically distributed (iid) data from a distribution with parameter θ , and let Y be a statistic formed from these data (e.g., $\bar{X}$). (Y is typically a sufficient statistic with a known distribution, but it need not be.) To determine an interval bound, we solve the following equation for θ :

$$F_Y(y_{\text{obs}} | \theta) - q = 0 ,$$

where

- $F_Y(\cdot)$ is the cumulative distribution function for Y
- y_{obs} is the observed statistic value
- q is an appropriate quantile

To determine the value for the appropriate quantile, we need to know two things: the type of interval we trying to construct (two-sided? one-sided lower bound? one-sided upper bound?), and whether the expected value of the adopted statistic, $E[Y]$, increases with θ (i.e., increases in value as θ increases in value), or decreases. Given those two pieces of information, we can pull the appropriate quantile value off of the following reference table

Interval Type	$E[Y]$ Increases With θ ?	q for Lower Bound	q for Upper Bound
two-sided	yes	$1 - \alpha/2$	$\alpha/2$
	no	$\alpha/2$	$1 - \alpha/2$
one-sided lower	yes	$1 - \alpha$	—
	no	α	—
one-sided upper	yes	—	α
	no	—	$1 - \alpha$

Example 1: An Analytic Solution

We sample a single datum with value $X = x_{\text{obs}} = 1$ from an exponential distribution with probability density function

$$f_X(x) = \frac{1}{\theta} e^{-x/\theta}.$$

What is a 95% lower-bound for θ ?

1. Identify an appropriate statistic. That's trivial here: $Y = X$ (and hence $y_{\text{obs}} = x_{\text{obs}} = 1$).
2. Determine the cdf for the random variable Y . Stated without proof, that's

$$F_Y(y|\theta) = 1 - e^{-y/\theta}.$$

3. The expected value of Y is $E[Y] = \theta$. As θ increases, $E[Y]$ increases.
4. We want a one-sided lower bound (with $\alpha = 0.05$) and, according to (3), we are on the “yes” line. Thus $q = 1 - \alpha = 0.95$.

We now have all the pieces necessary to derive the lower bound:

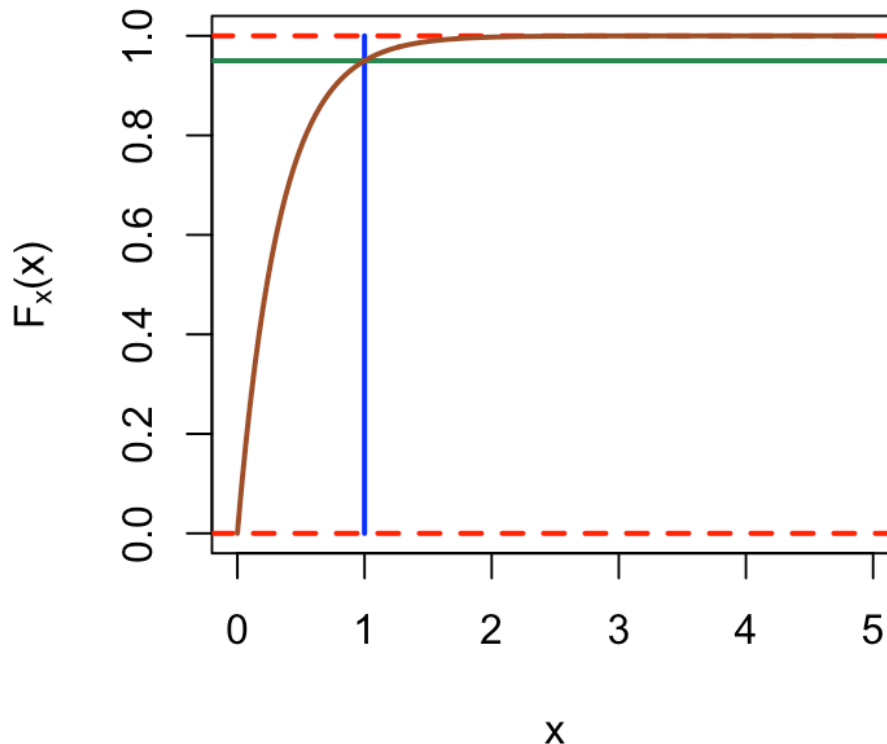
$$\begin{aligned} F_Y(y_{\text{obs}}|\hat{\theta}_L) - q &= 0 \\ \Rightarrow 1 - e^{-y_{\text{obs}}/\hat{\theta}_L} - 0.95 &= 0 \\ \Rightarrow e^{-y_{\text{obs}}/\hat{\theta}_L} &= 0.05 \\ \Rightarrow -y_{\text{obs}}/\hat{\theta}_L &= \log(0.05) \\ \Rightarrow \hat{\theta}_L &= -\frac{y_{\text{obs}}}{\log(0.05)} = 0.334. \end{aligned}$$

Below, we show how the cdf for the random variable Y , assuming $\hat{\theta}_L = 0.334$, passes through the intersection of the blue line (representing y_{obs}) and the green line (representing $q = 0.95$). Plots like this are good to show to students: if the observed value changes, the blue line shifts, and thus we have to change $\hat{\theta}_L$ to get the cdf to once again pass through the intersection of the lines. In short: confidence intervals are random intervals.

```

y.obs <- 1
plot(c(y.obs,y.obs),c(0,1),typ="l",xlab="x",ylab=expression(F[x]*"
(x)"),col="blue",xlim=c(0,5),lwd=2)
abline(h=0,col="red",lwd=2,lty=2)
abline(h=1,col="red",lwd=2,lty=2)
abline(h=0.95,col="seagreen",lwd=2)
theta <- 0.334
x.plot <- seq(0,5,by=0.01)
Fx.plot <- 1 - exp(-x.plot/theta)
lines(x.plot,Fx.plot,col="sienna",lwd=2)

```



Example 2: A Quasi-Analytic Solution

We sample $n = 10$ iid data from a normal distribution with mean μ and known variance $\sigma^2 = 4$. We observe $\bar{X} = 5$. What is a 95% upper bound on μ ?

1. Is $\bar{X}$ an appropriate statistic? Yes: it is a sufficient statistic (as found via likelihood factorization) and we know the sampling distribution for $\bar{X}$ (as found via the method of moment-generating functions):

$$\bar{X} \sim \square \left(\mu, \frac{\sigma^2}{n} \right) .$$

2. The cdf for the random variable $Y = \bar{X}$ is

$$F_Y(y|\mu, \sigma^2) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{y - \mu}{\sqrt{2}\sigma} \right) \right] .$$

3. We know that $E[Y] = E[\bar{X}] = \mu$, so $E[Y]$ does increase with μ .
4. We want a one-sided upper bound (with $\alpha = 0.05$) and, according to (3), we are on the “yes” line. Thus $q = \alpha = 0.05$.

So let's solve!

$$\begin{aligned} & F_Y(y_{\text{obs}}|\hat{\mu}_U, \sigma^2/n) - q = 0 \\ \Rightarrow & \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{\sqrt{n}(y_{\text{obs}} - \hat{\mu}_U)}{\sqrt{2}\sigma} \right) \right] - 0.05 = 0 \\ \Rightarrow & 1 + \operatorname{erf} \left(\frac{\sqrt{n}(y_{\text{obs}} - \hat{\mu}_U)}{\sqrt{2}\sigma} \right) = 0.1 \\ \Rightarrow & \operatorname{erf} \left(\frac{\sqrt{n}(y_{\text{obs}} - \hat{\mu}_U)}{\sqrt{2}\sigma} \right) = -0.9 \\ \Rightarrow & \frac{\sqrt{n}(y_{\text{obs}} - \hat{\mu}_U)}{\sqrt{2}\sigma} = \operatorname{erf}^{-1}(-0.9) \\ \Rightarrow & y_{\text{obs}} - \hat{\mu}_U = \sqrt{2} \frac{\sigma}{\sqrt{n}} \operatorname{erf}^{-1}(-0.9) \\ \Rightarrow & \hat{\mu}_U = y_{\text{obs}} - \sqrt{2} \frac{\sigma}{\sqrt{n}} \operatorname{erf}^{-1}(-0.9) = 6.040 . \end{aligned}$$

To compute the inverse error function, we use the `erfinv()` function of R's `pracma` package.

Does this match what we would derive using a canned formula from introductory statistics?

$$\bar{X} + z_{0.95} \frac{\sigma}{\sqrt{n}} = 5 + 1.645 \frac{2}{\sqrt{10}} = 6.040 ,$$

where $z_{0.95} = \text{qnorm}(0.95) = 1.645$. Yep...our result matches perfectly.

Example 3: A Numerical Solution

Let's repeat Example 2, but using R's `uniroot()` function.

First, we need to define a function that returns the value of $F_Y(y|\theta) - q$:

```
f <- function(mu,y.obs,q,sigma,n)
{
  pnorm(y.obs,mean=mu,sd=sigma/sqrt(n)) - q
}
```

Note that what we want to solve for (μ) has to be the first argument to `f`.

Second, we pass our new function to `uniroot()` along with a range of values over which to search for the one root of the equation:

```
uniroot(f,interval=c(-1000,1000),y.obs=5,q=0.05,sigma=2,n=10)$root
```

```
## [1] 6.040307
```

Done. Note that since μ can take on any value, we define a large range of negative and positive numbers over which to search. Also note that if σ^2 is unknown, we can utilize the t distribution and write, e.g.,

```
f <- function(mu,y.obs,q,S,n)
{
  pt((y.obs-mu)/(S/sqrt(n)),n-1) - q
}
uniroot(f,interval(-1000,1000),y.obs=5,q=0.05,S=[sample sd],n=10)
$root
```

where S is the sample standard deviation.

Example 4: But Does This Work with Discrete Distributions?

The short answer: yes. The key is that the parameter θ is, in typical situations, continuously valued, and so the basic algorithm doesn't change.

Let's conduct an experiment in which we flip a coin $k = 10$ times. We record the number of heads. We then repeat the experiment such that we have $n = 20$ outcomes, and we find that $\bar{X} = 116/20 = 5.8$. What is a two-sided confidence interval for the success probability p ?

1. It turns out that $\bar{X}$ is not an appropriate statistic here, because we cannot easily write down its sampling distribution. (We can define it exactly numerically, but...) On the other hand,

$$n\bar{X} = \sum_i X_i \sim \text{Binomial}(nk, p),$$

as one can easily derive using the method of moment-generating functions. Hence we'll use $Y = \sum_i X_i$, with the observed value $y_{\text{obs}} = 116$.

2. The cdf is given in (1).
3. We know that $E[Y] = nkp$ increases with p .
4. Thus the lower bound is associated with the value $q = 0.975$ and the upper bound is associated with the value $q = 0.025$.

Let's solve!

```
f <- function(p,y.obs,q,nk)
{
  pbinom(y.obs,size=nk,prob=p) - q
}
uniroot(f,interval=c(0,1),y.obs=116,nk=200,q=0.975)$root # lower bound
```

```
## [1] 0.5133761
```

```
uniroot(f,interval=c(0,1),y.obs=116,nk=200,q=0.025)$root # upper bound
```

```
## [1] 0.6492334
```

Hmm...it appears the coin may very well be an unfair one, as $p = 0.5$ falls outside the interval.

Example 5: What to Do When All Hope is Lost

AKA, Working with Beta Distributions

We sample $n = 5$ iid data from a $\text{Beta}(a, 2.6)$ distribution, with the observed data being $\{0.3313, 0.1908, 0.1089, 0.0006937, 0.1642\}$. What is a 95% one-sided upper bound for a ?

When we apply likelihood factorization, the sufficient statistic $U = \prod_{i=1}^n X_i$ pops out...but we don't necessarily want to use this in general computations, because products can blow up quickly. (Also, we don't know the sampling distribution for U .) Recalling that functions of sufficient statistics are themselves sufficient statistics, we define $Y = -\log U = -\sum_{i=1}^n \log X_i$. We still don't know the sampling distribution for this statistic, but it *is* a summation and thus numerically easier to work with. (And why the minus sign? So that the value of Y is a positive number...that's really the only reason.)

For a beta distribution, $E[X] = a/(a + b)$ increases as a increases (with b fixed), so...

- $E[\log(X)]$ also increases as a increases, so...
- $E[\sum_i \log(X_i)]$ also increases as a increases, so...
- $E[-\sum_i \log(X_i)]$ decreases as a increases.

This means that we are on the upper bound/“no” line of the reference table, and thus that $q = 1 - \alpha = 0.95$.

What now? We don't know the sampling distribution for Y . The jig is up. Game over. [Insert your own cliché here.]

What we can do is simulate the empirical cdf for Y , given a . We do this in the code chunk below. First, we define a function `f` inside of which we

1. create `num.sim` separate datasets, each of length `n`, given parameter values for `a` and `b`;
2. compute the statistic Y for each dataset;
3. determine the proportion of Y values that are less than `y.obs` (this is the empirical cdf $F_n(y_{\text{obs}} | a, b)$); and
4. return the value $F_n(y_{\text{obs}} | a, b) - q$.

We then pass `f` to `uniroot()` as we have before.

```
f <- function(a,n,b,y.obs,q,num.sim=1000000,seed=236)
{
  set.seed(seed)
  X <- matrix(rbeta(n*num.sim,shape1=a,
                    shape2=b),nrow=num.sim)
  Y <- apply(X,1,function(x){-sum(log(x))})
  sum(Y <= y.obs)/num.sim-q
}
# c(0.01,2) is a user-set range over which to explore
print(uniroot(f,c(0.01,2),n=5,b=2.6,y.obs=14.058,q=0.95)$root)
```

```
## [1] 0.9272251
```

We find that the approximate 95% upper bound on a is 0.927.

So: even when we *don't* know the sampling distribution of our adopted statistic, we can *still* estimate interval bounds! Huzzah!

We note that the default number of simulated datasets in the code above is 10^6 . This choice is driven a desire to make the result as precise as possible while keeping the runtime reasonable. (Here, the runtime is ≈ 1 CPU minute on a standard desktop or laptop computer.) The empirical cdf is an *estimate*, i.e., even if we hold a constant, $\text{sum}(Y \leq y.\text{obs})$ is a random variable. Let's denote the sum as $S|a$, the total number of simulated datasets as k , and the true cdf value (given y_{obs}) as p . Then

$$S|a \sim \text{Binomial}(k, p) .$$

The standard error for the empirical cdf is thus

$$\sqrt{V \left[\frac{1}{k}(S|a) \right]} = \sqrt{\frac{p(1-p)}{k}} .$$

If $k = 10^6$ and, e.g., $p = 0.025$, the standard error is $\sim 10^{-4}$, which is on par with typical uncertainties of estimates generated by `uniroot()`. We thus expect the uncertainties of our interval bound estimates to be $\sim 10^{-4}$.